SE 801 · SOFTWARE PROJECT LAB III · FINAL PROJECT, OCTOBER 2026

BanglaFactGuard

A multimodal Bangla fact-checking system: **source evidence** for headlines and photocards, **an AI model** for text + image posts, and **verdicts from weighted expert review**.

SUBMITTED BY
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GRAPHICAL ABSTRACT · FROM A SUSPICIOUS CLAIM TO A PUBLISHED VERDICT

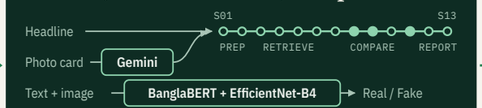
FINDING COLOURS · Found · matched · Altered · mismatched · Fake · Not found · incomplete

1 · INPUT
A suspicious Bangla claim



A headline (outlet, body and date optional), a photocard, or text with its image, from the website or the Chrome extension.

2 · AUTOMATED CHECK
Read it, find the source, compare



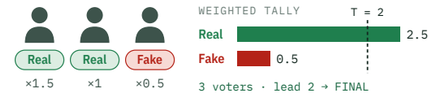
A background job searches registered Bangla outlets only, ranks the articles and compares the claim with the best match.

3 · PRELIMINARY FINDINGS
Three findings, not one label



Source, headline and date are reported separately and marked preliminary. No report found means 'Not found', never 'Fake'.

4 · EXPERT REVIEW
A credibility-weighted vote



Each vote carries a written reason and a weight from the expert's track record. No consensus by the limit → administrator.

5 · FINAL VERDICT
Published and delivered



Shown in the public Fact Explorer with every vote and reason; registered submitters get an in-app notice and an email.

REAL CASE Photo card submitted: "আমিন খানের সঙ্গে যে কেউ কাজ করতে চায়…" naming **prothomalo.com**

Gemini reads the card; the search finds the outlet's real report: "শাকিব খানের সঙ্গে যে কেউ কাজ করতে চায়…"

Source **Found** · Headline **Altered**, the person's name was swapped · Date **Mismatched** with the article

The experts could not agree within the review limit, so the claim was escalated to the administrator

Administrator's final decision: **Altered**. The report, with every vote and reason, is public in Fact Explorer

01 · WHY THIS PROJECT

Fake Bangla news borrows the look of real news

Misleading claims spread as posts, screenshots and news photocards that copy the look of trusted outlets.

1. **Convincing fakes.** Altered headlines, borrowed logos and templates, and old images make false claims look authentic.
2. **Slow manual checks.** A reader has to find the original report and compare its headline, outlet and date. Few people do.
3. **AI alone falls short.** A model can miss the right article, and a failed search proves nothing about a claim.

Missing: one Bangla tool that shows the evidence behind each finding and keeps people in charge of the verdict.

OBJECTIVES

01 Three ways to check headline, photo card, text + image	**02 Separate findings** source, headline and date
03 Preliminary AI Likely Real / Fake for text + image	**04 Human verdicts** weighted votes, admin escalation
05 Reuse and openness saved results, public archive	**06 Check while browsing** Chrome extension, admin tools

DESIGN PRINCIPLES

Not found ≠ Fake	Missing evidence is never a verdict.
Unavailable ≠ 0	Missing scores show as unavailable, not 0.
People decide	Findings stay preliminary until reviewed.
Known outlets	Evidence only from registered outlets.

Scope: Bangla text and images on the website and desktop Chrome. **Not covered:** other languages, video, social-media crawling, automated verdicts.

02 · METHODOLOGY A · SOURCE-BASED VERIFICATION

Find the source. Compare like with like.

Typed claims and claims read from photo cards share one 13-stage background pipeline.

PREPARE · RETRIEVE

- **S01 Normaliser** — cleans the text, sets the search scope
- **S02 Cache lookup** — compatible saved result? reuse it, skip ahead
- **S03 Query generator** — headline, keyword and date queries
- **S04 Source search** — outlet's own search and Google News
- **S05 Evidence retrieval** — httpx; Playwright when a page needs scripts
- **S06 Article extractor** — title, body and publication date
- **S07 Evidence ranker** — LaBSE + mMARCO MiniLM cross-encoder

RANKED ARTICLES

COMPARE · REPORT

- **S08 Source correspondence** — or Not found / Incomplete, never Fake · Found
- **S09 Headline alteration** — or no verdict when evidence is thin · Matched · Altered
- **S10 Body similarity** — TF-IDF · Jaccard · Levenshtein · LaBSE
- **S11 Date verification** — or Incomplete when a date is missing · Matched · Mismatched
- **S12 Result assembly** — safe defaults + a reason for each finding
- **S13 Persistence** — result, articles, queries; Redis pointer
- → **Expert review queue** — signed-in submitter is notified

INSIDE S09 · A REAL CASE: WHY "LOOKS SIMILAR" IS NOT ENOUGH

PHOTO CARD ~~আমিন খানের~~ সঙ্গে যে কেউ কাজ করতে চায়…
SOURCE TITLE শাকিব খানের সঙ্গে যে কেউ কাজ করতে চায়…

Meaning similarity · LaBSE	Contradiction · mDeBERTa-v3 NLI	Named person · BanglaTag NER
0.84 looks alike	**0.996**	**changed**

Headline · Altered — "The headline names আমিন খানের; the source title names শাকিব খানের instead."

Compared with the source **title** only, never the body. **Also checks** numbers (Bangla = English digits), dates, negation, people and organisations.

SEARCH SCOPE

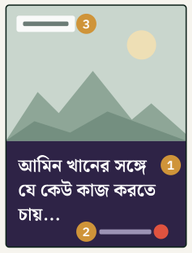
Outlet named → search it only. **Missing or unknown** → all 10 active registered outlets:

প্রথম আলো · কালের কণ্ঠ · যুগান্তর · সমকাল · ইত্তেফাক · মানবজমিন · বাংলাদেশ প্রতিদিন · নয়া দিগন্ত · ইনকিলাব · দ্য ডেইলি স্টার

03 · METHODOLOGY B · PHOTO CARDS AND TEXT + IMAGE

Two more ways in: photo cards and text + image

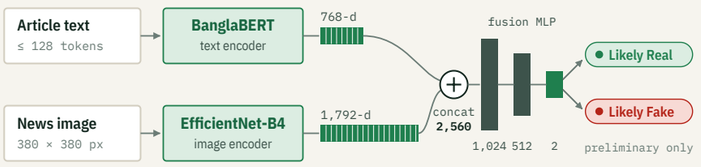
PHOTO CARD → CLAIM



1. Printed headline — আমিন খানের সঙ্গে যে কেউ কাজ করতে চায়…
2. Outlet name or logo — prothomalo.com → search scope
3. Printed date, if any — compared with the article date (S11)

Gemini vision reads all three in one pass and leaves a missing value empty instead of guessing. The claim then runs through S01–S13. An unreadable headline or a failed service stops the check with a clear reason. Photo cards have their own worker lane (up to 4 at a time), so retries never hold up other checks.

TEXT + IMAGE → PRELIMINARY PREDICTION



DATASET

MultiBanFakeDetect

9,600 Bangla text–image pairs, balanced Real / Fake

7,680 train · 960 val · 960 test

RECORDED RESULT

80% accuracy on 960 held-out pairs

	Prec.	Recall	F1
Real	0.79	0.81	0.80
Fake	0.81	0.79	0.80

TRAINING · REUSE

35 epochs · lr 2e-4 · weight decay 0.01 · batch 6 · label smoothing 0.1 · 3 warm-up epochs

Duplicate check: a saved prediction is reused only if text ≥ 0.92, image ≥ 0.85 and combined ≥ 0.90 cosine. Same text with a new image is a new check.

IN THE BROWSER · CHROME EXTENSION SIDE PANEL, ALL THREE METHODS

Use selected text · Capture + crop an area · Activity tab + alerts · Same account, own session

04 · METHODOLOGY C · EXPERT REVIEW

AI proposes. Experts decide.

1. **Pick a claim** from the review queue; an expert's own submissions never appear there.
2. **Read the evidence:** the claim, the preliminary findings and the source article.
3. **Vote** Real · Fake · Misleading · Altered with a justification of at least 50 characters.
4. **Weigh and re-check:** the vote is weighted and the decision rule is tested again.

VOTE WEIGHT BY CREDIBILITY TIER · ADMIN-EDITABLE

Novice	Competent	Expert	Master
×0.5	×1	×1.5	×2

0% · 40% · 70% · 90% · 100%

accuracy = share of an expert's past votes that matched the final verdict

New experts count ×1 until N = 10 completed reviews. Each keeps the weight it was cast with.

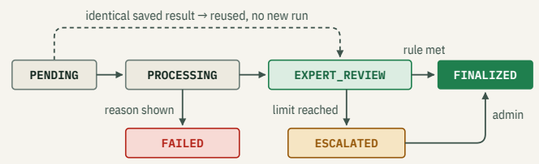
DECISION RULE

Final when `leader ≥ T` `voters ≥ M` `lead ≥ margin`

A tie never finalises. If the vote or time limit is reached first, the claim is **ESCALATED** and the administrator's decision is final.

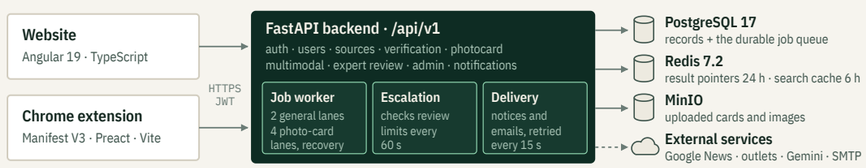
DEMO CONFIG · M=3 · T=2 · MARGIN=1 · LIMIT=3 VOTES

LIFE OF A SUBMISSION



05 · PLATFORM AND ARCHITECTURE

Two clients, one backend with in-process workers, three data stores



06 · WHO CAN DO WHAT

Four roles, one account system

FEATURE	GUEST	USER	EXPERT	ADMIN
Submit claims, all three methods	●	●	●	●
Fact Explorer and public reports	●	●	●	●
Chrome extension	●	●	●	●
My Submissions, notices, email		●	●	●
Review queue and weighted vote			●	
Escalations, experts, sources, rules				●

07 · TESTING AND RESULTS

Software tested, model measured

424 / 426 — **backend unit tests passed** in the recorded pytest run: 0 failed, 2 skipped (need local model weights), 20.4 s

69 — **documented test cases** (TC-01 to TC-69): access 14 · text & source 16 · photo card & text + image 15 · voting & admin 15 · integrity & security 9

82–153 s — **per source check** in 8 timed runs (pipeline only, before expert review); no formal load test yet

08 · NEXT STEPS

Limitations to address

1. **Measure the whole system** on labelled Bangla claims, not only the model.
2. **Harder photo cards:** unclear layouts, low-quality images.
3. **More evidence:** more outlets; sites that block fetching.
4. **Faster checks** and a concurrent load test.
5. **Public deployment** beyond the local setup.

Show the evidence. Let experts decide.

তথ্য যাচাই করুন, তারপর শেয়ার করুন

BUILT WITH

Angular 19 · FastAPI · SQLAlchemy · PyTorch · Hugging Face Transformers · sentence-transformers · Gemini API · PostgreSQL 17 · Redis 7.2 · MinIO · Preact + Vite (Chrome MV3) · pytest

Runs locally: website, API with background workers, PostgreSQL, and Redis + MinIO in Docker Compose.

SOURCE CODE

github.com/abulkalamazad1425/BanglaFactGuard